Week 6
Review

Applied Statistical Analysis/Quantitative Methods I

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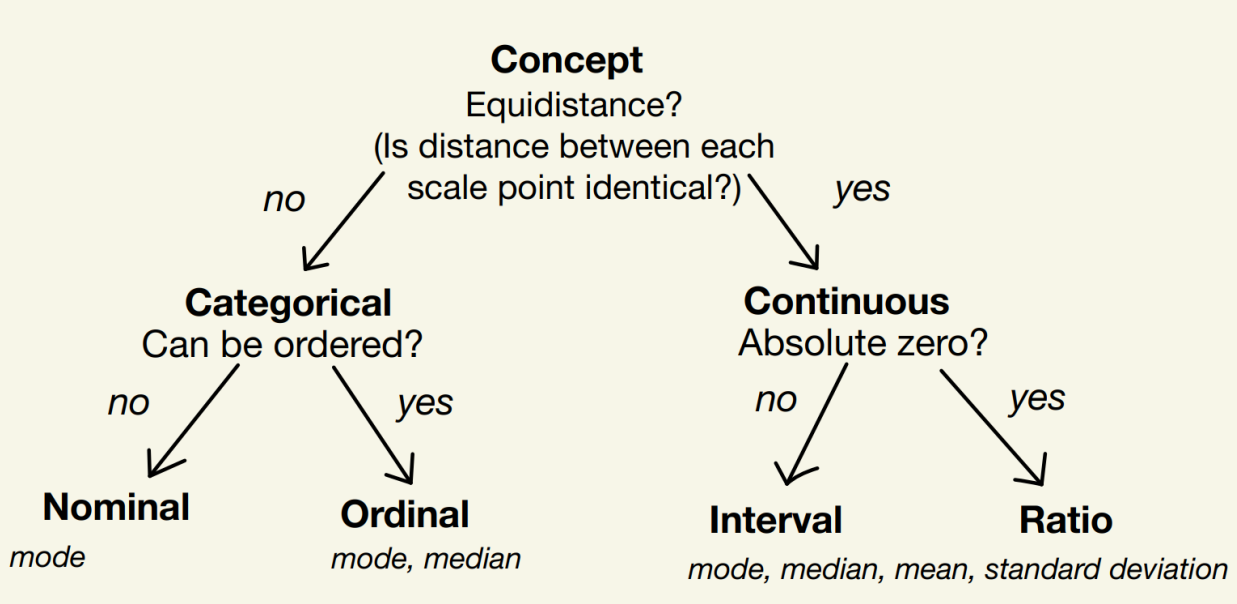
Autumn 2026

Week 1: Measurement Scales

Question

How can we measure concepts, and why does it matter?

- Discrete: a finite set of possible values
- Continuous: an infinite set of possible values



(Kellstedt and Whitten 2018, Chapter 5)

What's relationship b/w population & sample?

- Population: "total set of subjects of interest in a study"
- Parameter: "numerical summary of the population"
- Sample: "subset of population on which study collects data"
- Statistic: "a numerical summary of sample data"
- Observation: a single subject/unit, or one row in a dataset

(Agresti and Finlay 2009, 5)

Central Tendency and Variability

- Mean: Sum of all values divided by number of observations

$$\bar{y} = \frac{1}{n} \sum_{i=1}^{n} y_i$$

- Variance: Sum of squared deviations divided by $n - 1$, where a deviation is $y_i - \bar{y}$:

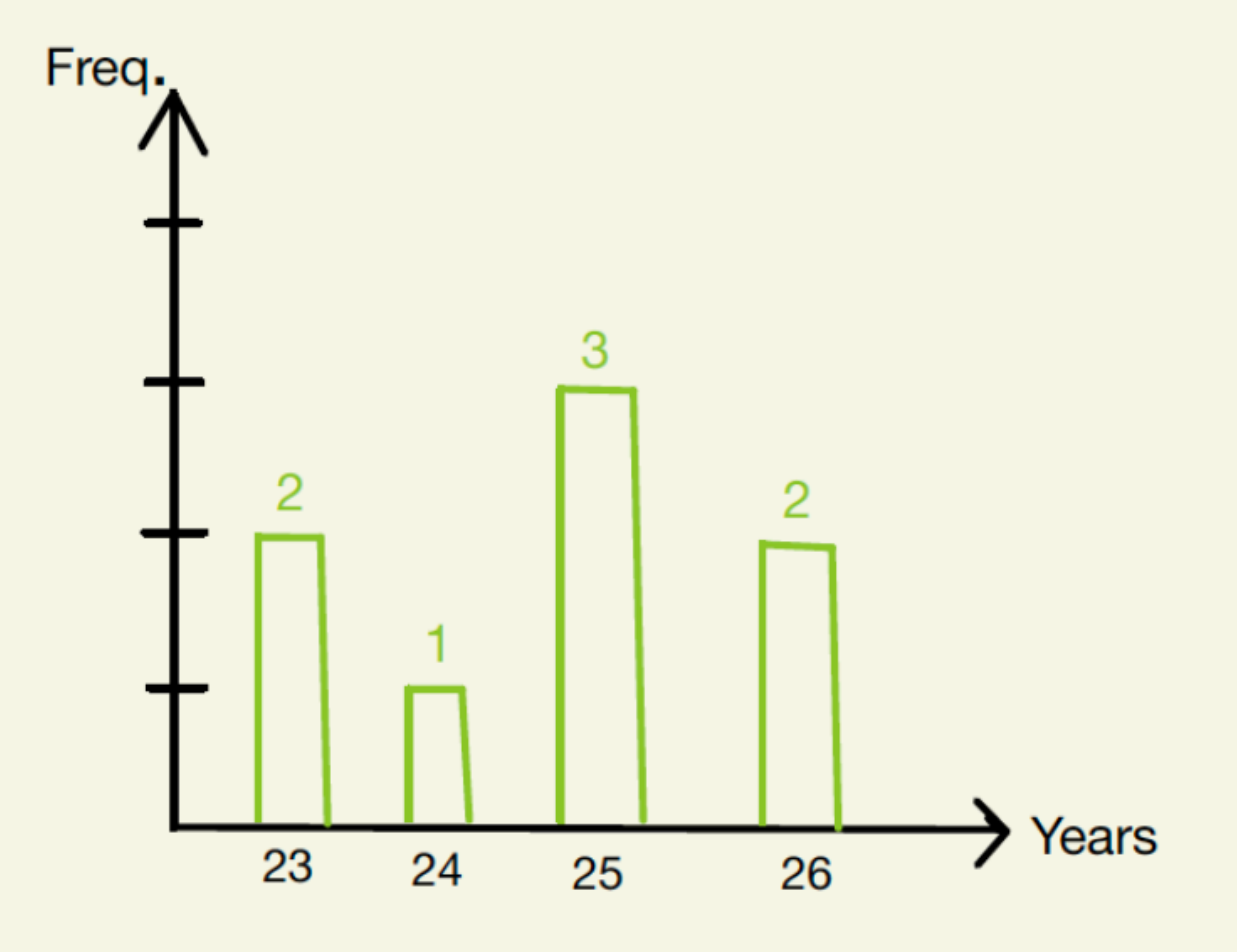
$$s^2(y) = \frac{\sum_{i=1}^{n} (y_i - \bar{y})^2}{n - 1}$$

- Standard Deviation: Square root of the variance, returning measure to original units:

$$s = \sqrt{\frac{\sum_{i=1}^{n} (y_i - \bar{y})^2}{n - 1}}$$

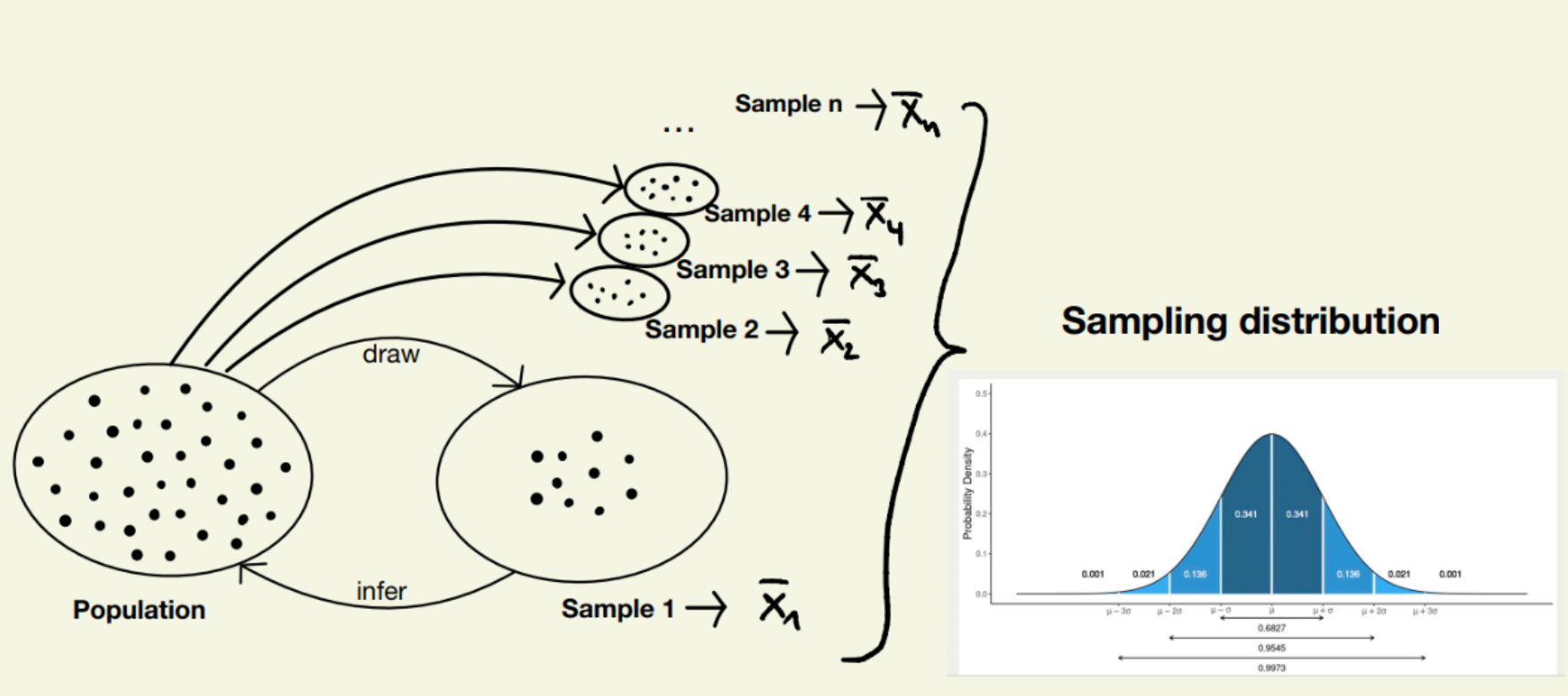
Distributions and Probability Distributions

A probability distribution “lists the possible outcomes and their probabilities” (Agresti and Finlay 2009, 75)



Sampling Distribution

Theoretically…



Sampling Distribution

A sampling distribution of a statistic is probability distribution that specifies probabilities for possible values statistic can take (Agresti and Finlay 2009, 87)

In other words

A probability distribution for a statistic rather than for individual observations. Ex: What is probability of $\bar{Y} = 0.5$, rather than the probability of $Y = 3$?

Sampling Distribution: Why It Matters

- Probability theory helps us predict how close a statistic falls to parameter it estimates (Agresti and Finlay 2009, 87)
  - How close is $\bar{y}$ to μ?
- Usually, we have only one sample and one estimate
  - A point estimate is "a single number that is best guess for parameter value" (Agresti and Finlay 2009, 107)

Sampling Distribution of Mean

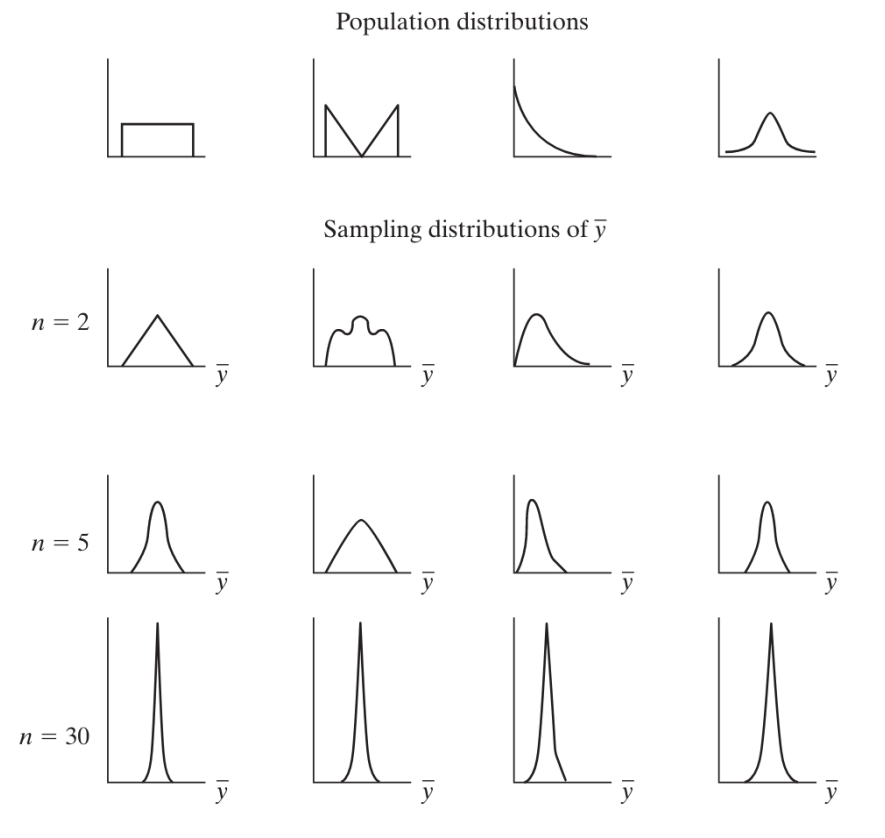
- If we repeatedly took samples, then in long run, mean of sample means would equal population mean μ (Agresti and Finlay 2009, 90)
  - Thus, mean of sampling distribution of $\bar{y}$ equals μ
- Standard error describes how much $\bar{y}$ varies from sample to sample (Agresti and Finlay 2009, 90):

$$\sigma_{\bar{y}} = \frac{\sigma}{\sqrt{n}}$$

- Why does this work?

What is the Central Limit Theorem

For random sampling with a large sample size n, sampling distribution of sample mean $\bar{y}$ is approximately normal, regardless of population distribution



(Agresti and Finlay 2009, 93)

Central Limit Theorem

Knowing that sampling distribution of $\bar{y}$ can be approximated by a normal distribution helps us find probabilities for possible values of $\bar{y}$ (Agresti and Finlay 2009, 94)

Why does it matter?

This is key to inferential statistics!!!

Confidence Intervals

- Confidence Interval: Interval of numbers around point estimate that we believe contains parameter value (Agresti and Finlay 2009, 110):

$$\text{Point estimate} \pm \text{Margin of error}$$

- Confidence Level: Probability that method produces an interval containing parameter (usually 0.95 or 0.99) (Agresti and Finlay 2009, 110)
- Margin of error: Is a multiple of standard error

$$\sigma_{\bar{y}} = \frac{\sigma}{\sqrt{n}}$$

- At 95% confidence level, margin of error is $\pm 1.96 \sigma_{\bar{y}}$

Week 2: Null-Hypothesis Significance Testing

TABLE 6.1: The Five Parts of a Statistical Significance Test

1. **Assumptions**
   Type of data, randomization, population distribution, sample size condition
2. **Hypotheses**
   Null hypothesis, H_0 (parameter value for "no effect")
   Alternative hypothesis, H_a (alternative parameter values)
3. **Test statistic**
   Compares point estimate to H_0 parameter value
4. ***P*-value**
   Weight of evidence against H_0; smaller P is stronger evidence
5. **Conclusion**
   Report P-value
   Formal decision (optional; see Section 6.4)

(Agresti and Finlay 2009, 147)

Significance Test for a Mean

TABLE 6.3: The Five Parts of Significance Tests for Population Means

1. **Assumptions**
   Quantitative variable
   Randomization
   Normal population (robust, especially for two-sided H_a, large n)
2. **Hypotheses**
   $H_0: \mu = \mu_0$
   $H_a: \mu \neq \mu_0$ (or $H_a: \mu > \mu_0$ or $H_a: \mu < \mu_0$)
3. **Test statistic**
   $t = \dfrac{\bar{y} - \mu_0}{se}$ where $se = \dfrac{s}{\sqrt{n}}$
4. ***P*-value**
   In t curve, use
   P = Two-tail probability for $H_a: \mu \neq \mu_0$
   P = Probability to right of observed t-value for $H_a: \mu > \mu_0$
   P = Probability to left of observed t-value for $H_a: \mu < \mu_0$
5. **Conclusion**
   Report P-value. Smaller P provides stronger evidence against H_0 and supporting H_a. Can reject H_0 if $P \leq \alpha$-level.

(Agresti and Finlay 2009, 155)

Significance Test for a Difference in Means

- **Null and alternative hypotheses (Step 2):**

$$H_0 : \bar{y}_1 = \bar{y}_2 \quad \text{or} \quad \bar{y}_1 - \bar{y}_2 = 0$$

$$H_a : \bar{y}_1 \neq \bar{y}_2$$

- **Test statistic (Step 3):** measures number of standard errors between estimate and the H_0 value (Agresti and Finlay 2009, 192):

$$t = \frac{\text{Estimate of parameter} - \text{Null hypothesis value of parameter}}{\text{Standard error of estimate}}$$

$$t = \frac{(\bar{y}_1 - \bar{y}_2) - 0}{\text{SE}}, \qquad \text{SE} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

- **Null and alternative hypotheses:**

$$H_o : \text{two variables are independent}, \qquad f_o = f_e$$

$$H_a : \text{two variables are dependent}, \qquad f_o \neq f_e$$

- Test statistic compares observed contingency-table frequencies with those expected under independence (Agresti and Finlay 2009, 225):

$$X^2 = \sum \frac{(f_o - f_e)^2}{f_e}$$

Correlation

- Covariance is average product of deviations of two quantitative variables from their means:

$$\text{cov}(X, Y) = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{n}$$

- Correlation (Pearson's r) is standardized covariance:

$$r_{xy} = \frac{\text{cov}(X, Y)}{S_X S_Y}$$

It ranges from -1 to 1

Testing the Significance of Correlation

- **Null and alternative hypotheses:**

$$H_0 : \rho_{xy} = 0 \qquad \text{(no association between } X \text{ and } Y\text{)}$$

$$H_a : \rho_{xy} \neq 0 \qquad \text{(an association between } X \text{ and } Y\text{)}$$

- Test statistic:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

Ordinary Least Squares (OLS)

- How do we find line which "best" fits data?
- Apply OLS, which minimizes sum of squared errors (SSE)
- SSE is sum of squared differences between actual and predicted values of Y

$$\begin{aligned}\text{SSE} &= \sum_{i=1}^{n} \hat{\epsilon}_i^2 \\ &= \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2 \\ &= \sum_{i=1}^{n} \Big(Y_i - (\hat{\alpha} + \hat{\beta} X_i)\Big)^2\end{aligned}$$

We want to minimize this quantity

Assumptions of Linear Regression: Errors

$$Y_i = \alpha + \beta X_i + \epsilon_i, \qquad \epsilon_i \sim N(0, \sigma^2)$$

- ϵ_i is normally distributed (needed for inference)
- $E(\epsilon_i) = 0$: no bias
  - If this is violated, error is not random
  - May be correlated with an omitted variable
- ϵ_i has constant variance σ^2 (homoscedasticity; contrasted with heteroscedasticity)
- Parametric linearity: $Y_i = \alpha + \beta X_i + \epsilon_i$

(Kellstedt and Whitten 2018, 190–194)

Requirements of Linear Regression

- X must vary, why?
- Number of observations must be larger than number of predictors, why?
- In multiple regression, there must be no perfect multicollinearity, why?

(Kellstedt and Whitten 2018, 190–194)

Inference About the Slope

- **Null and alternative hypotheses:**

$$H_0 : \beta = 0 \qquad \text{(no association between } X \text{ and } Y\text{)}$$

$$H_a : \beta \neq 0 \qquad \text{(an association between } X \text{ and } Y\text{)}$$

- Test statistic measures number of standard errors between estimate and null-hypothesis value (Agresti and Finlay 2009, 192):

$$t = \frac{\text{Estimate of parameter} - \text{Null hypothesis value of parameter}}{\text{Standard error of estimate}}$$

$$t = \frac{\hat{\beta} - \beta_{H_0}}{\text{SE}(\hat{\beta})} = \frac{\hat{\beta}}{\text{SE}(\hat{\beta})}, \qquad H_0 : \beta = 0$$

Question

How can we partition error?

1. Total Sum of Squares (SS_T)
   - Total sum of squares is sum of squared differences between observed values of Y and mean:

$$\mathrm{SS_T} = \sum_{i=1}^{n} (y_i - \bar{y})^2$$

Partitioning the Error

2. Residual sum of squares is sum of squared differences between observed values of Y and the regression line:

$$\mathrm{SS_R} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

3. Model sum of squares is sum of squared differences between regression line and mean of Y:

$$\mathrm{SS_M} = \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2$$

Regression Analysis: Model Performance

- R^2: Proportion of variation in Y explained by X
- Ranges from 0 to 1
- If X explains all variation in Y, then $R^2 = 1$

$$\text{SST} = \text{SSM} + \text{SSR}, \qquad \text{SSM} = \text{SST} - \text{SSR}$$

$$R^2 = 1 - \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{Variation not explained by model}}{\text{Total variation in } Y}$$

$$R^2 = \frac{\text{SSM}}{\text{SST}} = \frac{\text{Variation explained by model}}{\text{Total variation in } Y}$$

Class business

- Read required (and suggested) online materials
- Finish Problem set #2!